

Benchmarking Generative AI and Physics-Based Molecular Simulation for Sampling Conformational Heterogeneity in T4 Lysozyme

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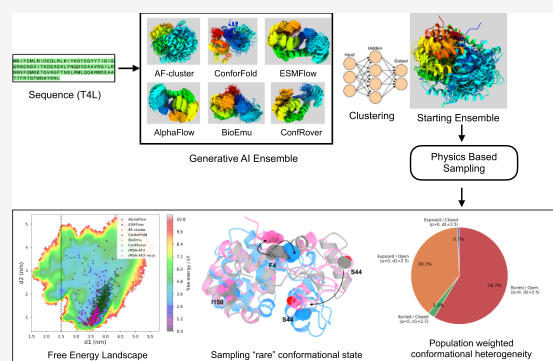


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ABSTRACT: Wild-type T4 lysozyme (T4L) is used as a benchmark to evaluate conformational sampling across generative AI, AI-accelerated molecular simulation (AMS), and physics-based enhanced molecular dynamics (EMD). A four-state model—exposed/open, exposed/closed, buried/open, and buried/closed; is defined using physically meaningful collective variables. Generative AI methods (AF-cluster, MSA subsampling of AlphaFold2, ConforFold, AlphaFlow, ESMFlow, Conforover, BioEmu) largely sample only the exposed/open state. In contrast, AMS integrates generative ensembles with physics-based molecular simulations, recovering all states and reproducing equilibrium populations similar to EMD and experimental smFRET signatures.



INTRODUCTION

Proteins are molecular machines that govern biological function. They are not static structures but interconvert between multiple transiently populated metastable states, each capable of activating distinct signaling pathways. The populations of these states are modulated by environmental perturbations such as temperature, pH, ligand binding, and mutations. Sampling the transiently populated states is therefore central to understanding how proteins encode function, yet it remains one of the hardest problems in structural biology.^{1–5}

The development of generative AI (GenAI) algorithms such as AlphaFold has enabled accurate prediction of static protein structures from sequence alone, but these algorithms do not sample transiently populated alternate conformational states. A growing subfield of machine learning for structural biology has aimed to address this by predicting conformational ensembles from sequence alone.¹ These approaches fall into two subgroups: MSA-based methods, which perturb evolutionary coupling information in multiple sequence alignments to generate structural diversity, and generative models such as diffusion models or flow-matching frameworks, trained on short MD simulations data, which claim to sample transiently populated alternate states. Despite rapid growth in both subgroups, systematic benchmarking against physically rigorous reference data remains absent, making it difficult to assess whether these methods genuinely capture biologically relevant conformational diversity or simply reproduce structural noise.

Physics-based enhanced sampling methods such as metadynamics,^{6–9} adaptive sampling,¹⁰ and replica exchange¹¹ provide such a reference. They can sample transiently populated

alternate states and their relative populations across a diverse range of biomolecular systems. Sampling efficiency is assessed by how well these methods capture state populations along collective variables (CVs) that discriminate transient states from the ground state.¹² However, fair comparisons between physics-based enhanced sampling and GenAI methods are rare due to three key challenges. A minimal benchmark system must have experimentally validated transient states with CVs simple enough to distinguish those states from the ground state. It must have sufficient structural data in the Protein Data Bank so that limited training data cannot be used as a confounding factor. And system size must not unfairly disadvantage GenAI methods, which often struggle with larger and more conformationally complex systems.

Wild-type T4 lysozyme (T4L) satisfies all three criteria. It is small (164 residues), has single-molecule FRET (smFRET) data that directly validates transiently populated alternate states, and has been extensively studied by physics-based enhanced sampling methods. Yet, no prior study has systematically compared GenAI methods, AI-accelerated molecular simulation, and physics-based enhanced sampling on the same system using well characterized CVs. I close this gap here. Additionally,

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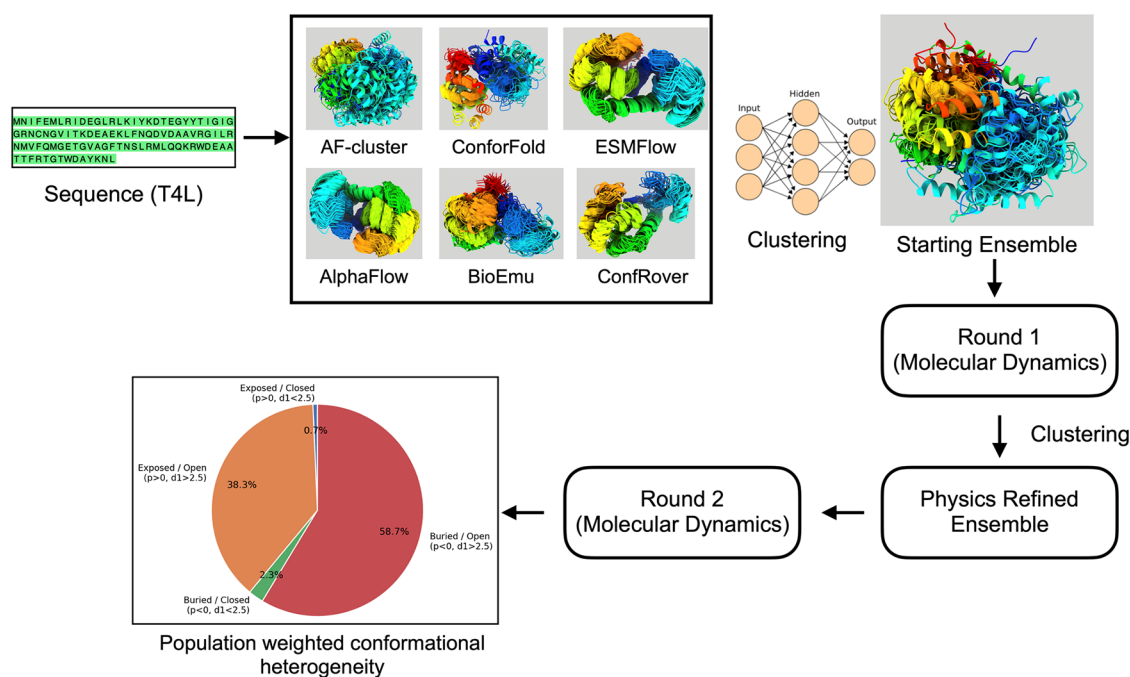


Figure 1. Schematic of the AI-accelerated molecular simulation (AMS) workflow. The workflow begins with the T4L sequence, followed by ensemble generation using multiple GenAI methods, including AF-cluster, ConforFold, BioEmu, ConfRover, ESMFlow, and AlphaFlow. K-center clustering was then performed using transformed distance features to select an initial “starting ensemble” of $N = 100$ structures. Unbiased molecular dynamics simulations of 200 ns were initiated from each member of this starting ensemble in Round 1. The resulting trajectories were clustered using the K-center algorithm on transformed distances, yielding a “physics-refined ensemble” of $N = 80$ structures. Additional unbiased molecular dynamics simulations of 200 ns each were then initiated from the physics-refined ensemble in Round 2. A Markov state model (MSM) was constructed using RMSD features from the combined Round 1 and Round 2 simulations. Equilibrium populations were then projected onto different collective variables to predict conformational populations and heterogeneity in T4L.

I introduce a new framework called AI-accelerated molecular simulation (AMS). AMS combines multiple GenAI methods to construct a highly diverse structural prior, which is then used to seed independent molecular dynamics simulations and capture the full scope of conformational dynamics in T4L (Figure 1). By uniting these generative models into a single bucket of ensemble generator, AMS overcomes the sampling deficiencies inherent to any individual algorithm. This approach yields a significantly broader initial distribution and increases the probability of capturing rare conformational events compared to relying on a single GenAI protein ensemble generator augmented by molecular simulations.

RESULTS

I used two sets of CVs to describe the conformational ensemble of T4L. The first is a two-dimensional CV defined by two Ca distances: $d1$ (Ser44–Ile150) and $d2$ (Glu22–Gln141). The Ser44–Ile150 residue pair was used by Miller and co-workers¹³ to drive metadynamics simulations, enabling comparison of predicted smFRET signals with experimental smFRET data to probe the opening and closing motion of T4, with $d1 < 2.5$ nm defining the closed state and $d1 > 2.5$ nm defining the open state. The Glu22–Gln141 distance, introduced by Abou-Hatab and Abrams,¹⁴ serves as an orthogonal CV capturing hinge domain motion. The second CV is the locking coordinate p , developed by Stock and co-workers,¹⁵ which quantifies the solvent exposure of Phe4 relative to the hinge helix. A positive value ($p > 0$) indicates that Phe4 is solvent-exposed, while a negative value ($p < 0$) indicates it is buried within the interdomain interface. Combined with $d1$, this defines a four-state model: exposed/closed, exposed/open, buried/closed, and buried/open. This

four-state classification enables quantitative comparison of state populations across all methods.

In this paper, I benchmarked three classes of methods. The first class comprises GenAI approaches, divided into two subgroups. The first subgroup consists of MSA-based methods: AF-cluster,¹⁶ which clusters MSA subsamples as input to AlphaFold2;¹⁷ reduced MSA subsampling with AlphaFold2 (rMSA-AF2);¹⁸ and ConforFold,¹⁹ a retrained OpenFold2 model designed to predict alternate conformations. The second subgroup consists of generative models trained on structural or simulation data: AlphaFlow,²⁰ ESMFlow,²⁰ ConfRover,²¹ and BioEmu.²² The second class is AMS, a methodological contribution of this work, in which GenAI ensembles seed multiple rounds of unbiased MD simulations that are combined using a Markov state model (MSM)^{23,24} to recover physically refined state populations (Figure 1). The third class is physics-based enhanced sampling repurposed from Miller and co-workers,¹³ combining 5 μ s unbiased MD simulations with metadynamics-seeded unbiased MD simulations (hereafter enhanced molecular dynamics, EMD), which serves as the quantitative reference for all comparisons.

AMS successfully sampled all four conformational states: exposed/open, exposed/closed, buried/open, and buried/closed. The two dominant states, exposed/open (AMS: 38.3%, EMD: 39.1%) and buried/open (AMS: 58.7%, EMD: 54.0%), were recovered with relative populations comparable to EMD. The transiently populated buried/closed state was also recovered with a comparable population (AMS: 2.3%, EMD: 2.9%). The exposed/closed state, however, was underrepresented in AMS relative to EMD (AMS: 0.7%, EMD: 3.9%), suggesting that this particular transient state remains the most

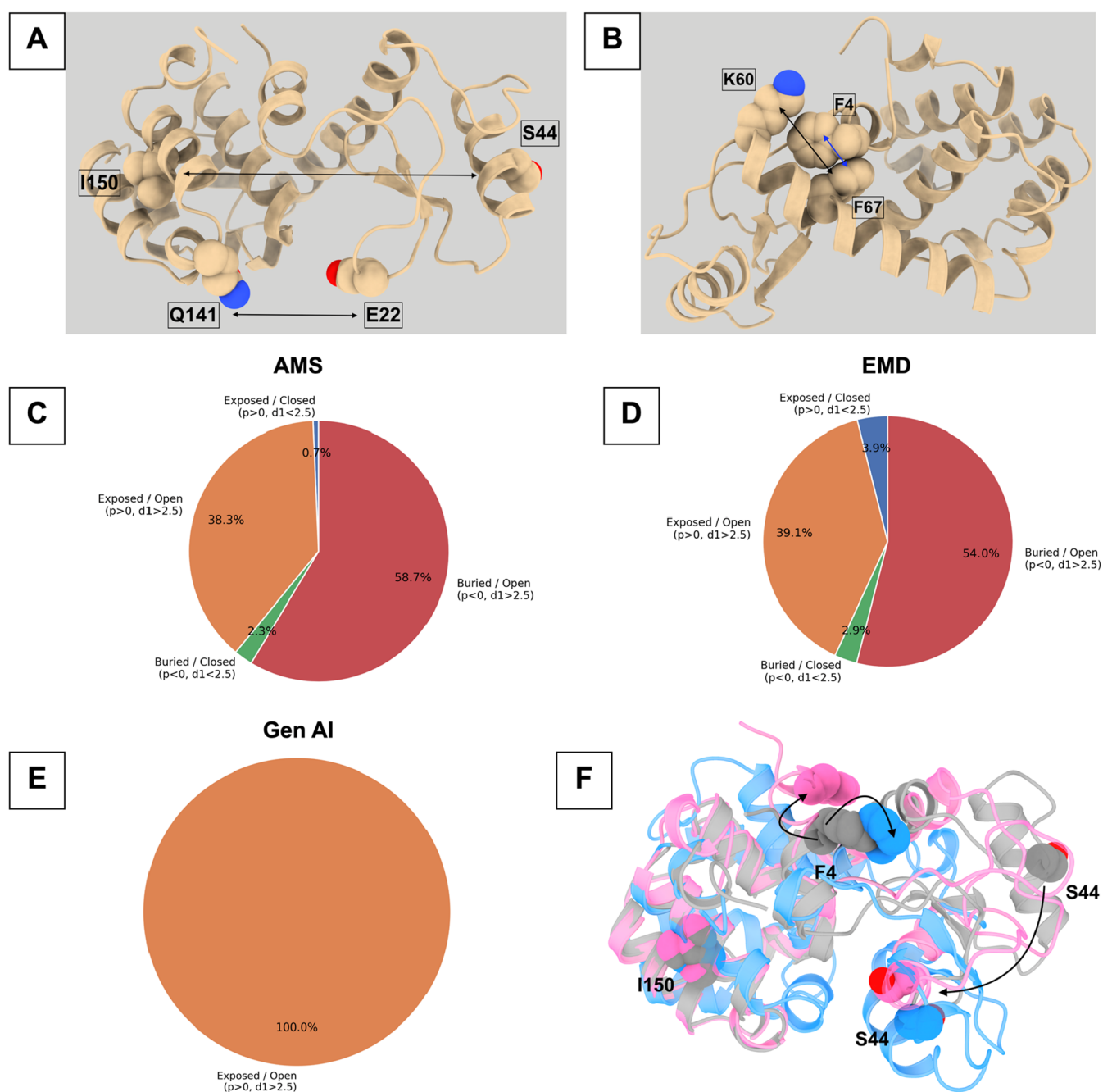


Figure 2. Collective variables used to characterize T4 lysozyme conformational heterogeneity and comparison of sampling across different approaches. (A) Pictorial representation of the $d1$ and $d2$ collective variables, defined as the $C\alpha$ – $C\alpha$ distances between Ser44 (S44) and Ile150 (I150) and between Glu22 (E22) and Gln141 (Q141), respectively. The $d1$ coordinate distinguishes closed and open conformations, with $d1 < 2.5$ nm corresponding to the closed state and $d1 > 2.5$ nm corresponding to the open state. (B) Definition of the locking coordinate p , which describes the position of the Phe4 (F4) side chain relative to the hydrophobic cavity. The cavity direction is defined by the $C\alpha$ atoms of Lys60 and Phe67, while the position of Phe4 is represented by the center of mass of the carbon atoms in its phenyl ring. Positive values of p indicate that the Phe4 side chain is solvent-exposed, whereas negative values indicate that it is buried within the hydrophobic cavity. (C–E) Comparison of conformational sampling by AMS, EMD, and GenAI ensembles alone. AMS and EMD sample the full spectrum of conformational heterogeneity, whereas the combined GenAI ensembles predominantly sample a single state. (F) Superposition of crystal T4 lysozyme structure shown in dark gray with representative exposed/closed and buried/closed conformations shown in magenta and blue, respectively. The comparison highlights conformational changes involving Ser44 and Phe4, which contribute to the observed conformational heterogeneity.

challenging to access without explicit enhanced sampling to seed follow up unbiased simulations (Figure 2 and 3, see Figures S6 and S7 in Supporting Information for error analysis). It is important to highlight that populations derived from AMS converge robustly across choice of lag times and cluster centers (Figures S2 and S3 in Supporting Information). In contrast,

EMD suffers from disconnected transition matrices even when carefully seeded with short metadynamics runs specifically chosen to bridge open and closed states and create a fully connected MSM (Figure S8 in Supporting Information).

In contrast, GenAI ensembles alone remained predominantly trapped in the exposed/open state, failing to access the

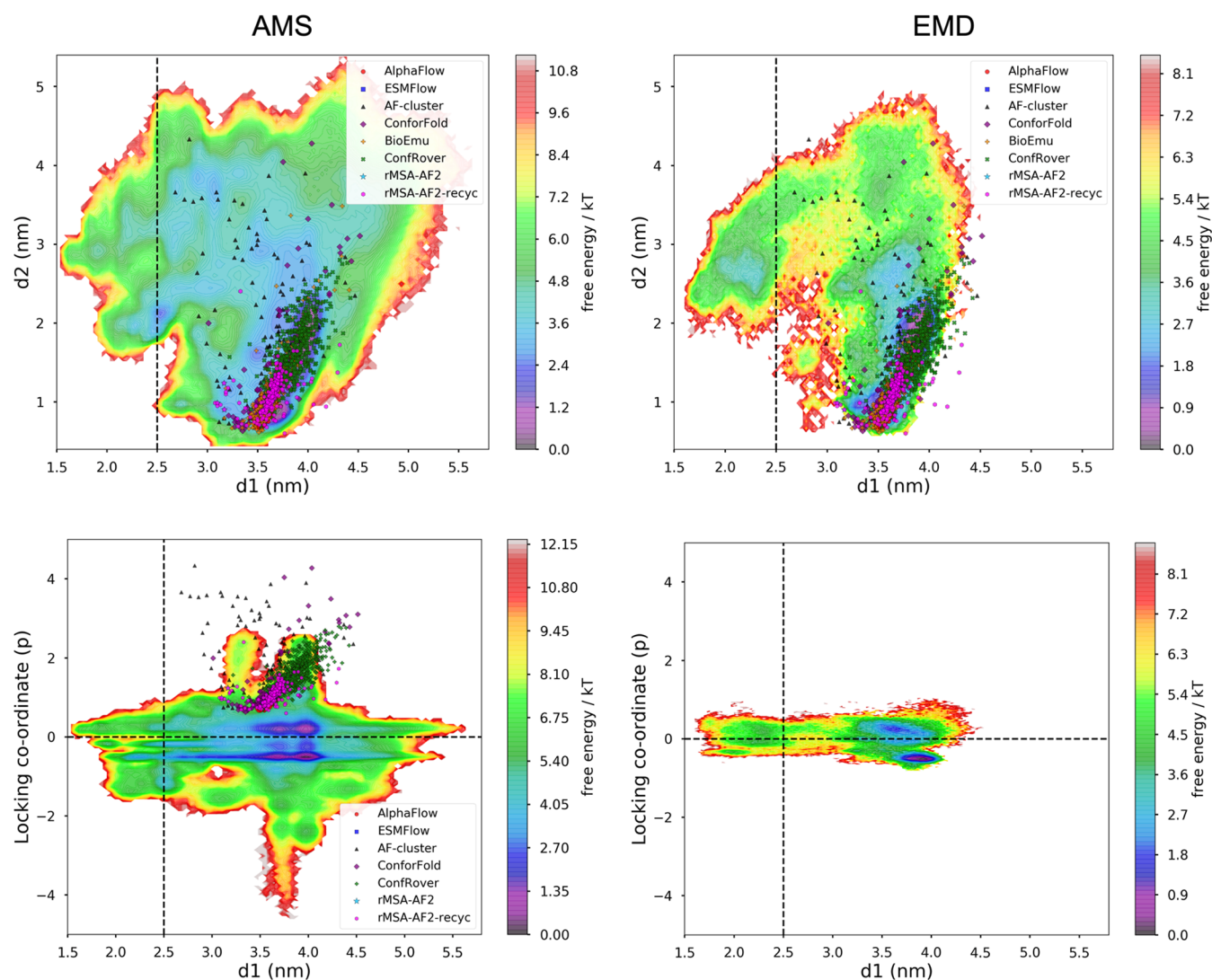


Figure 3. MSM-weighted free-energy landscapes projected onto different collective variables compare conformational sampling by the AMS and EMD approaches. The upper panels show equilibrium populations projected along the $d1$ and $d2$ collective variables, indicating that both AMS and EMD sample the closed state of T4 lysozyme, defined by $d1 < 2.5$ nm. The black dotted line marks the closed–open boundary at $d1 = 2.5$ nm. The lower panels show the projection of the locking coordinate p along $d1$, demonstrating that both AMS and EMD sample conformations with exposed and buried Phenylalanine 4 side-chain states, corresponding to $p > 0$ and $p < 0$, respectively, across both closed and open T4 lysozyme conformations. Data points from the GenAI ensembles are projected onto the physics-refined free-energy surfaces to assess the conformational coverage of each method. Among the GenAI approaches, AF-cluster samples a substantially broader conformational landscape than the other methods.

transiently populated closed and buried states. Interestingly, unbiased MD simulations (20 independent 1 μ s simulations initiated from the T4L X-ray crystal structure, PDB: 5LZM) sampled conformational transitions between the exposed/open and buried/open states (Figure 4). This indicates that although GenAI ensembles capture conformational heterogeneity along distance variables ($d1$ and $d2$), they alone cannot sample the buried/open state; suggesting that GenAI itself does not capture the underlying physics of the conformational transitions in T4L.

Projection of GenAI ensembles onto the $d1$ and $d2$ CV space revealed that AF-cluster provided substantially broader conformational coverage than all other GenAI methods (Figure 3). This suggests that the enhanced sampling efficiency of AMS is predominantly driven by the ability of AF-cluster to populate high-energy intermediate structures that serve as productive seeds for subsequent MD simulation. To test this hypothesis directly, I repeated the AMS protocol with AF-cluster excluded from the initial ensemble. The resulting simulations failed to

recover the conformational heterogeneity captured by the full AMS protocol. However, it still sampled a broader free energy surface when compared to unbiased MD simulations (Figures 4 and 5).

To further validate AMS against experiment, I computed the MSM-weighted smFRET distance distribution for the Ser44–Ile150 pair and compared it directly with experimental measurements (Figure S1 in Supporting Information). AMS recovered the transiently populated closed state in quantitative agreement with experiment. This result is significant because the EMD simulations of Miller and co-workers¹³ were explicitly seeded with the objective of sampling the closed state using enhanced sampling, yet AMS achieved comparable recovery of this rare state starting from unbiased MD alone. The ability to reproduce an experimentally observable transiently populated conformation without any explicitly specific target state demonstrates that AMS can sample rare conformational events that are inaccessible to GenAI methods alone.

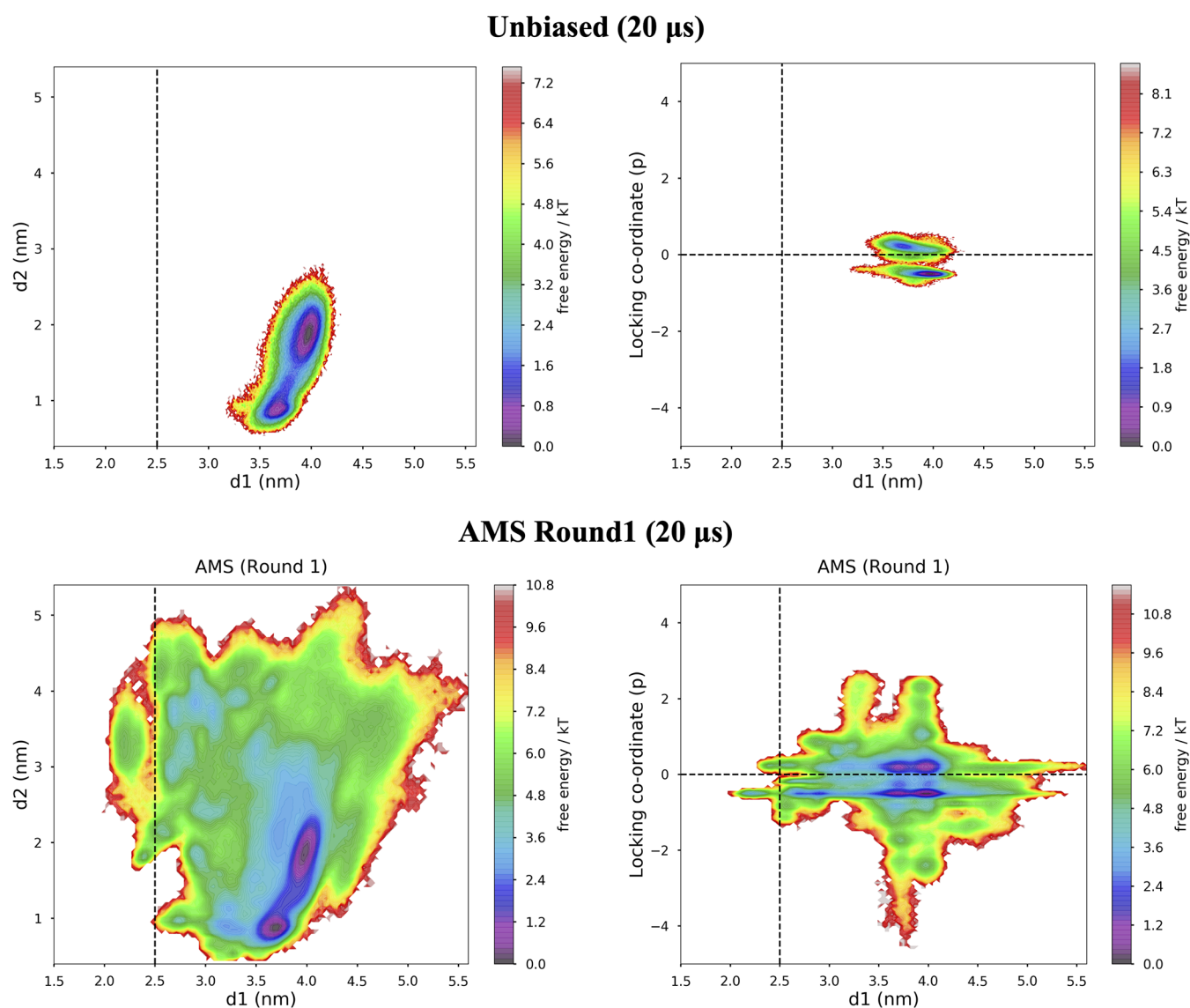


Figure 4. Free energy surfaces projected along $d1$ vs $d2$ and along $d2$ vs p (the locking coordinate) from a combined 20 μ s unbiased molecular dynamics data set (20 independent MD simulations, 1000 ns each), initiated from PDB structure 1LZM (following the approach of Miller *et al.*), are compared against AMS round 1 alone (100 $\times$ 200 ns each = total 20 μ s). This comparison highlights the efficiency of AMS over conventional unbiased MD simulations starting from a single starting structure in capturing conformational heterogeneity in wild-type T4 lysozyme.

CONCLUSIONS

This work introduces T4L as a minimal community benchmark for evaluating GenAI methods that claim to capture conformational heterogeneity from sequence alone. T4L is uniquely suited to this role: its experimentally validated transient states, well characterized CVs, extensive structural coverage in the Protein Data Bank and availability of enhanced molecular dynamics simulations data set collectively eliminate the confounding factors that have hampered fair comparisons in the field. The four-state classification framework presented here provides a transferable, physically interpretable scoring scheme that can be applied consistently across any method that generates structural ensembles of T4L.

The results presented here draw a clear boundary between what GenAI can and cannot do in its current form in context of T4L. Majority of the GenAI methods alone reproduce the dominant ground state (except AF-cluster which samples multiple high energy transient states) but fail to sample transiently populated closed state in T4L and their relative

populations. AMS, by using GenAI ensembles as seeds rather than as end products, recovers the full conformational landscape with populations that are quantitatively comparable to those from physics-based enhanced sampling. Two rounds of unbiased MD simulation, initiated from GenAI derived starting structures, were sufficient to achieve this. In comparison, unbiased MD simulations of the same length as round 1 of AMS starting from the X-ray crystal structure of wild type fails to sample the closed state in T4L (Figure 4) [Supporting Information](#). This is a practically important result: AMS requires no knowledge of the target state and no enhanced sampling bias, yet it matches the EMD's population percentages for the exposed/open, buried/open, and buried/closed states for wild type T4L; despite EMD being explicitly designed to find the closed state. Further, the seeding strategy that combines all the conformational ensembles generated by different GenAI algorithms alleviates the limitations of each one of them and provides a much wider distribution of conformations as initial

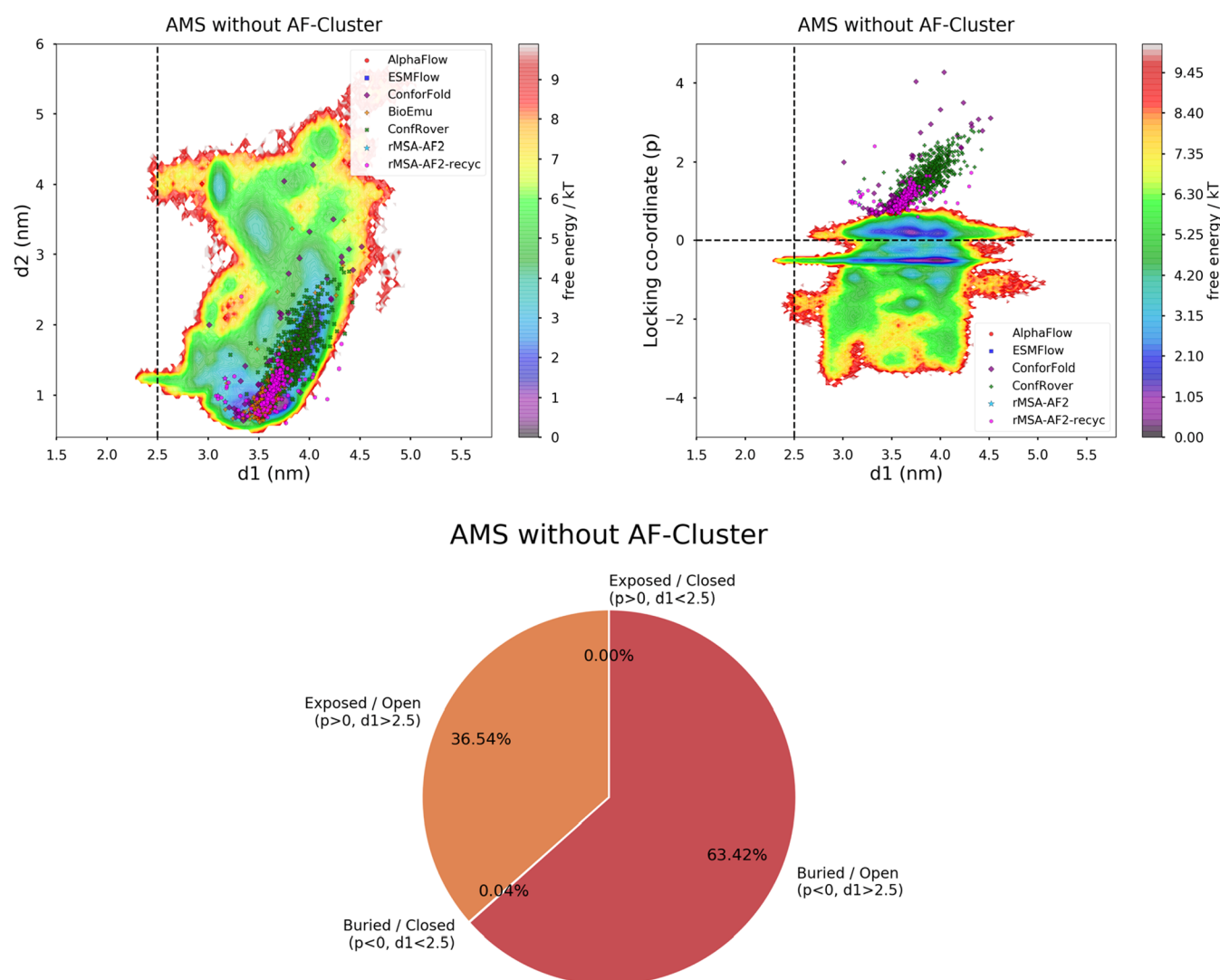


Figure 5. MSM-weighted free energy surfaces projected along two sets of collective variables show that molecular simulations initiated from the GenAI ensemble without the AF-cluster fail to sample the closed state. In contrast, AMS simulations that include the AF-cluster successfully capture this state, as shown in Figures 2 and 3. To generate the simulation data set, 100 cluster centers were selected from a combined conformational ensemble comprising AlphaFlow, ESMFlow, ConforFold, BioEmu, ConfRover, and rMSA-AF2 structures. For each cluster center, a 200 ns molecular dynamics simulation was performed, yielding a total of 20 μ s of simulation data.

seeds, which accelerates sampling when combined with physics-based molecular simulations.

The success of AMS, and of all generative-AI-seeded molecular simulation approaches such as AlphaFold-RAVE,²⁵ AlphaFold-SFA,²⁶ AlphaFold-MSM,²⁷ and BioEmu augmented molecular simulations,²⁸ depends heavily on the diversity of the initial seed. However, the dependency of GenAI augmented simulations on initial seeds has not been explicitly explored by previous literatures. In this case, I have clearly shown that AMS without structures from AF-cluster fails to sample the closed state of T4L. However, one encouraging factor is that even though these simulations failed to sample the closed state, they still sampled a wider conformational landscape than unbiased MD simulations. One can perform *K*-center²⁹ clustering on the combined simulation data to generate a diverse seeding ensemble for launching a second round of simulations. Figure S9 in the Supporting Information shows the data points corresponding to *K*-center clustering on top of AMS simulations run without AF-cluster, where one can see that short MD simulations launched from these seeds would most likely sample

the closed state of T4L. This strategy of seeding the next round of MD simulations from GenAI augmented molecular simulations can be used in the same spirit as adaptive sampling to explore the conformational landscape of biomolecules whenever the initial round of sampling fails to recover alternate states or needs additional data to build fully connected MSM.

Further, MSM generated from the EMD protocol developed by Miller *et al.*¹³ failed to form a fully connected MSM, unlike AMS round 1 (Figure S8 in Supporting Information). It is also worth highlighting that EMD alone (termed limited sampling by the authors) could not sample the minor state of T4L (closed state) in the context of the smFRET distribution (Figure 3A in Miller *et al.*¹³). The authors therefore needed to run additional adaptive sampling simulations using FAST¹⁰ to build a fully connected MSM and reach reasonable agreement with the experimental MSM. However, it is not possible to reproduce the results of Miller *et al.* for a fair, direct comparison, because the authors did not share the predicted FRET data needed for recalculation. In our protocol, AMS alone was able to sample this alternate state.

It is also worth noting that experimental FRET efficiency and simulated FRET efficiency are not expected to match exactly. Simulated FRET efficiency depends strongly on the number of cluster centers, the choice of feature sets used to build the MSM, and the underlying uncertainty introduced by the choice of force field and water model. Even so, it is encouraging that AMS was able to sample the minor state corresponding to the closed population of T4L, without requiring the more involved workflow used by Miller *et al.*: unbiased MD, followed by metadynamics, followed by further unbiased MD from metadynamics-seeded trajectories, followed by FAST adaptive sampling and finally connected all the data using MSM. Future work is in progress to examine how the choice of force field, water models and feature space affects the agreement with experimental data from smFRET, NMR, and other biophysical techniques.

It is worth mentioning that outside the two macrostate regions (open and closed) highlighted in Figure S1 (see Supporting Information), the AMS derived FRET efficiency distribution shows regions of divergence from the experimental smFRET signature. These regions correspond to transitional conformations that are not resolved as distinct free-energy minima and are not the focus of the macrostate-level comparison presented here. EMD does show a similar closed-state population, as reported by Miller *et al.*; however, exact population weighted comparison of AMS vs EMD is not possible due to the lack of available raw data from EMD simulations.

AMS also compares favorably to existing simulation strategies in terms of practicality and accessibility. In principle, AMS is a moderate-scale unbiased molecular dynamics simulation when compared to Folding@Home.³⁰ However, access to Folding@Home is restricted to a small number of researchers and requires a specialized software interface to orchestrate the simulations. Enhanced sampling methods such as metadynamics can also sample conformational heterogeneity; but requires setting up numerous hyperparameters such as Gaussian height, width, bias factor, and a reasonable choice of CV, each of which demands expert judgment and system-specific tuning.^{6–8} A poorly chosen CV can also silently bias the resulting free energy landscape in ways that are difficult to detect after the fact. Weighted ensemble methods are likewise large-scale and unbiased molecular dynamics simulations but carry their own significant implementation nuances. Among these approaches, AMS is the most straightforward to launch and use it requires no additional hyperparameter tuning or specialized expertise. It also does not manipulate the underlying potential energy surface as enhanced sampling methods such as metadynamics,⁶ Gaussian accelerated molecular dynamics (GaMD),³¹ or replica exchange³² do. Any research group with access to an HPC cluster equipped with only a few GPUs can perform AMS, making it one of the most practical choices available for sampling conformational heterogeneity in biomolecules. As an example, completing Round 1 of AMS for T4L (100 independent simulations of 200 ns length) requires only about 10 days of wall time on a lightweight compute node equipped with three NVIDIA A100 GPUs (~650 ns/day per GPU). AMS also provides a straightforward way to gain intuition about a reasonable CV. A typical practice to build such intuition is to perform multiple short and independent unbiased MD simulations from different starting points, but this only works well if the starting points are already diverse. MD simulations from a single starting structure remain trapped in local conformational minimum and yields a CV that reflects only

local, not global dynamics. This limitation can be addressed by training any CV discovery algorithm¹² of the user's choice (such as PCA, TICA, SFA, or autoencoders) on the combined ensemble from GenAI methods, or by running simulations shorter than AMS round 1 (~50 ns each) and combining them to train such an algorithm. The resulting CV can then be used with OneOPES³³ or a traditional metadynamics-like bias potential to enhance sampling and capture conformational heterogeneity across a wide range of biological systems.

Overall framework of the AMS protocol is readily transferable for sampling conformational heterogeneity across different protein classes including kinases, proteases, GPCRs, TNF-superfamily members where protein dynamics modulates downstream signaling pathways implicated in human diseases.

Looking forward, T4L and the benchmarking framework established here are well-positioned to evaluate the next generation of hybrid methods. These include GenAI seeded weighted ensemble simulations,³⁴ inference-time enhanced sampling approaches such as Boltz-MetaDiffusion,³⁵ and related methods that either combine GenAI with physics-based sampling protocols or bias the generative process during inference to access alternate states. The framework is equally applicable to MSA subsampling strategies applied to newer versions of foundation models such as AlphaFold3³⁶ and OpenFold3,³⁷ where the impact of richer training data and improved architecture on conformational diversity remains an open question. As these methods mature, rigorous benchmarking against experimentally validated populations along physically meaningful CVs will be essential to distinguish genuine advances in conformational sampling from improvements in ground-state structure prediction. T4L, with its tractable size, rich experimental data, and well-defined conformational heterogeneity, makes it a minimal benchmark system for that purpose.

METHODS

Ensemble Generation

Structural ensembles of T4 lysozyme (T4L) were generated from the wild-type T4L sequence using multiple GenAI based approaches, as summarized in Table 1. The input amino acid sequence used for all methods was:

Table 1. GenAI Methods Used for T4L Ensemble Generation, Number of Generated Structures, and Corresponding Codebase or Implementation

methods	number of structures	codebase/implementation
AF-cluster	132	https://github.com/HWaymentSteele/AF_Cluster
AlphaFlow	500	https://github.com/bjing2016/alphaflow (model: AlphaFlow-MD)
ConforFold	600	https://github.com/strauchlab/ConforFold
ConfRover	500	https://github.com/ByteDance-Seed/ConfRover
ESMFlow	100	https://github.com/bjing2016/alphaflow (model: ESMFlow-MD)
BioEmu	499	https://github.com/microsoft/bioemu
rMSA-AF2	80	Colabfold (MSA:8:16, num_recycles = 6)
rMSA-AF2-recyc	285	AlphaFold2_advanced_v2 (MSA:8:16,num_recycles = 3)

MNIFEMLRIDEGRLRLKIYKDTEGYTYI-
GIGHLLTKSPSLNAAKSELDKAIGRNCNGVITK-
DEAEKLFNQDVAARGILRNAKLKPVYDSLDAVRRCA-
LINMVFQMGETGVAGFTNSLRMLQQRWDEAAVNLAISR-
WYNQTPNRAKRVITTFRTGTWDAYKNL

The generated ensembles included both multiple-sequence-alignment-based approaches and generative models trained to produce conformationally diverse protein structures. For each method, the number of generated structures and the corresponding codebase or implementation are listed in Table 1. All structures generated across these methods were pooled into a combined GenAI ensemble for subsequent transformed-distance analysis, dimensionality reduction, and clustering.

Definitions of Collective Variables

The opening and closing motion of T4 lysozyme was characterized using the $\text{Ca}-\text{Ca}$ distance between Ser44 and Ile150, denoted as d_1 . Conformations with $d_1 < 2.5$ nm were classified as closed, whereas conformations with $d_1 > 2.5$ nm were classified as open. A second $\text{Ca}-\text{Ca}$ distance, measured between Glu22 and Gln141 and denoted as d_2 , was used as an orthogonal projection coordinate to describe hinge-domain motion, following the collective-variable definition proposed by Abou-Hatab and Abrams.¹⁴

The solvent exposure of Phe4 was quantified using the locking coordinate, p . For each trajectory frame, p was calculated by projecting the vector from the Ca atom of Phe67 to the center of the Phe4 phenyl ring onto the vector connecting the Ca atoms of Lys60 and Phe67. The center of the Phe4 phenyl ring was defined as the geometric center of the CG, CD1, CD2, CE1, CE2, and CZ carbon atoms. The locking coordinate was computed as

$$p = \frac{\mathbf{d}_{60,67} \cdot \mathbf{d}_{67,4}}{|\mathbf{d}_{60,67}|^2} \quad (1)$$

where $\mathbf{d}_{60,67}$ is the vector from the Lys60 Ca atom to the Phe67 Ca atom, and $\mathbf{d}_{67,4}$ is the vector from the Phe67 Ca atom to the center of the Phe4 phenyl ring. Positive values of p ($p > 0$) indicate that the Phe4 side chain is solvent-exposed, whereas negative values indicate that the side chain is buried ($p < 0$) within the hydrophobic cavity.

Feature Selection, Dimensionality Reduction and Clustering

To construct a distance-based representation of conformational variability, structures generated by the different GenAI methods were first pooled into a single all-atom ensemble. For every structure in this combined ensemble, all $\text{Ca}-\text{Ca}$ pairwise distances were converted into continuous contact-like variables using a smooth switching function, following Bhakat *et al.*³⁸

$$S(r) = \frac{1}{1 + \left(\frac{r}{r_{\text{decay}}}\right)^n} \quad (2)$$

Here, r denotes the $\text{Ca}-\text{Ca}$ distance, r_{decay} defines the distance scale over which the contact changes from formed to broken, and n controls the steepness of this transition. Distances much shorter than r_{decay} produce values close to 1, corresponding to a formed contact, whereas distances much longer than r_{decay} produce values close to 0, corresponding to a broken contact. This transformation replaces raw distances with a smooth and noise-tolerant measure of contact strength. I used $r_{\text{decay}} = 0.90$ nm and $n = 6$.

The same distance-to-contact transformation was used at two stages of the workflow. First, it was applied to the combined GenAI ensemble to identify structurally diverse starting conformations for molecular simulation. Second, after the unbiased molecular dynamics simulations were completed, the resulting trajectories were processed using the same transformed-distance representation to identify conformations for the physics-refined ensemble. Thus, both the initial GenAI predicted structures and the MD-derived conformations were analyzed in a common feature space.

For each $\text{Ca}-\text{Ca}$ pair, the transformed contact values were evaluated across the relevant structural data set, either the pooled GenAI ensemble or the postprocessed simulation trajectories. Contacts were retained only if they showed evidence of both formation and disruption, according to the criterion:

$$\min(S) < 0.4 \text{ and } \max(S) > 0.6 \quad (3)$$

This filtering step removes distance pairs that remain essentially unchanged, either always formed or always broken, and keeps only contacts that report meaningful conformational rearrangements. In this representation, values of $S(r) \geq 0.6$ correspond to formed contacts, values of $S(r) \leq 0.4$ correspond to broken contacts, and intermediate values describe partially formed or fluctuating contacts.

The filtered set of transformed distance features was then used for dimensionality reduction with Slow Feature Analysis (SFA), as introduced for biomolecular simulation data by Vats *et al.*²⁶ SFA identifies linear combinations of input features that change slowly over time, thereby emphasizing collective motions associated with long-time scale conformational rearrangements. For an input signal $c(t)$, SFA generates output coordinates $y_k(t) = g_k(c(t))$ by minimizing temporal variation:

$$\Delta y_k = \langle \dot{y}_k^2 \rangle \quad (4)$$

Applied to the transformed $\text{Ca}-\text{Ca}$ contact features, SFA identified the slowest collective distance patterns that distinguish conformationally heterogeneous regions of T4 lysozyme. The first two slow features were used as the reduced feature space for K-center clustering.²⁹ Cluster centers ($k = 100$) selected from the combined GenAI ensemble were saved as PDB structures and used as starting points for the first round of molecular dynamics simulations. The same procedure was subsequently applied during postprocessing of the simulation trajectories to select the physics-refined ensemble for the next round of simulations. SFA and K-center clustering were performed using the MDML software package (GitHub: <https://github.com/svats73/mdml/tree/main>)

Molecular Dynamics Simulations

Each selected structure from starting ensemble or physics-refined ensemble was prepared using the *tleap* module in Amber2022,^{39,40} following the general simulation protocol described by Meller *et al.*²⁷ Protein atoms were parametrized with the AMBER *ff14SB*⁴¹ force field. Each system was neutralized with the appropriate counterions and solvated in a truncated-octahedron box of TIP3P water molecules, with a minimum distance of 10 Å between any protein atom and the edge of the simulation box.

Energy minimization was carried out in two steps. First, solvent molecules and ions were minimized while harmonic restraints were applied to the protein atoms using a force constant of 100 kcal mol⁻¹ Å⁻². This was followed by an

unrestrained minimization of the full solvated system. The resulting *Amber* topology and coordinate files were then converted to *GROMACS* format using *Acpype*,⁴² and all subsequent equilibration and production simulations were performed using *GROMACS* 2022.⁴³

Systems were gradually heated from 0 to 300 K over 500 ps in the NVT ensemble while applying positional restraints to backbone heavy atoms with a force constant of 500 kJ mol⁻¹ nm⁻². After heating, each system was equilibrated for 200 ps in the NPT ensemble at 300 K and 1 bar without positional restraints. Temperature was maintained using the velocity-rescale thermostat, and pressure was controlled using the *Parrinello–Rahman* barostat.⁴⁴

Unbiased production simulations were then performed in the NPT ensemble using a 2 fs integration time step. Nonbonded interactions were treated with a 1.0 nm cutoff, long-range electrostatics were calculated using the particle-mesh Ewald method,⁴⁵ and bonds involving hydrogen atoms were constrained using the LINCS⁴⁶ algorithm. Production trajectories were initiated from the SFA selected cluster centers and ran for 200 ns each, with coordinates saved every 10 ps.

Enhanced Molecular Dynamics Simulations

The molecular dynamics simulations used in this study were adapted from *Miller et al.*¹³ Briefly, the data set comprises a 5 μ s unbiased MD trajectory, which did not sample the open-to-closed conformational transition. To drive the transition, the authors applied a metadynamics bias along the α – α distance between Ser44 and Ile150, extracted snapshots along the resulting free-energy pathway, and launched short unbiased MD simulations from each structural snapshot. Rather than analyzing the metadynamics trajectories directly, I combined the primary 5 μ s unbiased trajectory with a series of short, unbiased simulations seeded along the biased pathway, yielding a total simulation time of 27.6 μ s. This combined data set included 17 simulations (1 μ s each) initiated from alternate structures and 56 simulations (0.1 μ s each) initiated from cluster centers extracted from the unbiased trajectory.

Markov State Model

Markov state models (MSMs) were constructed using the *PyEMMA*⁴⁷ package with RMSD-based featurization of both AMS and EMD trajectories. K-means clustering ($K = 200$) was employed to discretize conformational space, and state populations were computed across multiple lag times to assess convergence and compare the relative equilibrium distributions of AMS and EMD ensembles (Figures S2 and S5 in Supporting Information). Figure S3 and S4 in the Supporting Information shows the populations of T4L conformational states as a function of the number of cluster centers, demonstrating the apparent convergence of the AMS simulations.

smFRET Prediction

smFRET prediction was carried out using *Enspara*⁴⁸ software package (<https://enspara.readthedocs.io/en/latest/smFRET.html>). Trajectories from AMS were featurized by transformed distances followed by SFA (as described in “Feature selection, dimensionality reduction and clustering” subsection) followed by K-means clustering with $k = 500$. smFRET was predicted using the protocol described by *Miller et al.*¹³

■ ASSOCIATED CONTENT

Data Availability Statement

GenAI ensembles, closed states of T4L and codes to calculate the distance, RMSD and locking coordinate can be accessed here: [10.5281/zenodo.20111229](https://doi.org/10.5281/zenodo.20111229), https://osf.io/jcxbs/overview?view_only=2c9e9b57afe84521bf80950412f0747a (OSF entry with Dropbox link which contains RMSD, distance as well as smFRET analyses and corresponding metadata) and <https://github.com/sbhakat/Gen-AI-t4l-bechmarking>

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.6c02044>.

Conformational state definitions and comparison with experiment: MSM-weighted FRET efficiency distributions from AMS benchmarked against experimental smFRET, showing sampling of both the open and closed states of T4 lysozyme, including the rare closed-state population previously reported only via EMD simulations (Figure S1); convergence and robustness analysis: lag-time convergence and MSM-weighted free-energy surfaces for AMS and EMD, equilibrium population stability across cluster number (K) and feature choice (RMSD-based vs TICA-based), Bayesian MSM-weighted equilibrium populations with 95% confidence intervals for the four conformational states in both AMS and EMD, and transition count-matrix connectivity analysis comparing network connectivity between AMS and EMD (Figures S2–S8); and conformational sampling diversity and seeding strategy: K-center clustering of AMS-sampled conformations on the free-energy surface, illustrating the diversity of states used to seed round 2 of AMS sampling of the closed state (Figure S9) (PDF)

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Author Contributions

S.B. designed the study, performed all calculations and analyses, and wrote the manuscript.

Notes

The author declares no competing financial interest.

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